

Female on Board: Causal Evidence on Firm Performance

Mariia Dorfman

Abstract

We study whether firms with female representation on their board of directors perform better. Using the UK firms' dataset and exploiting the 2015 recommendation to increase female representation on boards, we estimate the causal effect of female directors on firm performance. We apply several CIA-based estimators including OLS with controls, propensity score matching, IPWRA, and coarsened exact matching. After accounting for observable firm characteristics and ensuring covariate balance, we find no statistically significant effect of female board representation on return on assets (ROA). Robustness checks using alternative outcomes and placebo tests confirm the stability of the results.

Introduction

This paper examines whether female representation on corporate boards affects firm performance. We study UK firms around the 2015 recommendation encouraging firms to appoint more female directors. Identifying this effect is challenging because board composition is not randomly assigned: firms with female directors may differ systematically from other firms in size, governance, and baseline performance. To address this issue, we use selection-on-observables methods, including OLS with controls, propensity score matching, inverse probability weighted regression adjustment (IPWRA), and coarsened exact matching. Across all specifications, we find no robust evidence that female board representation improves firm performance, measured by return on assets.

Data

The dataset contains firm-level observations for UK companies between 1999 and 2019. All firms had no female directors prior to 2010. The treatment occurs after the 2015 recommendation encouraging firms to appoint female board members.

We clean and prepare the data in several steps. First, to avoid anticipation effects, we exclude the years 2011–2014, as the recommendation was announced in 2011 and firms may have adjusted behavior in advance. Second, we drop observations with missing values for the treatment indicator or ROA. The number of such observations is small and does not materially affect the sample. Third, we winsorize all continuous variables at the 1st and 99th percentiles to reduce the influence of outliers.

Our main outcome is firm performance measured by return on assets (ROA). We compute average ROA over 2015–2019 to capture post-treatment performance and reduce noise in yearly outcomes. As a robustness check, we also consider the change in ROA, defined as the difference between average ROA in 2015–2019 and ROA in 2010.

We convert the data into a cross-sectional format with one observation per firm. All covariates are measured in 2010 to ensure they are predetermined with respect to treatment. We include firm-level controls capturing baseline performance and size (ROA, revenue, gross margin, capital expenditures, total assets, market capitalization, and employment), as well as board characteristics (board size, share of independent directors, directors' network size, average age, and qualifications). Industry fixed effects based on TRBC classifications are also included.

While the dataset contains a larger set of potential covariates, many of these variables have substantial missing values. Including a large number of controls would significantly reduce the available sample and complicate the matching procedure by increasing the dimensionality of the covariate space. We therefore restrict the analysis to a set of 12 key firm and board characteristics that we consider most relevant for predicting both treatment assignment and firm performance. The exact variables are described in Table 1.

Missing values in covariates are replaced with medians, which are less sensitive to outliers and allow us to retain observations without distorting distributions.

Finally, we compare treated and control firms and find substantial differences in observable characteristics prior to matching, indicating strong selection into treatment. Table 1 presents summary statistics.

Table 1: Summary Statistics by Treatment Group with Raw Differences (p-values)

	Control				Treated				p-value
	Mean	SD	Min	Max	Mean	SD	Min	Max	
ROA Post (2015–2019)	-13.859	30.423	-152.848	30.788	-0.403	21.643	-126.091	37.518	<0.001
Change in ROA	-3.433	36.361	-124.487	149.421	-2.736	23.149	-114.406	159.500	<0.001
ROA 2010	-10.425	34.875	-152.848	37.518	2.333	25.617	-152.013	37.518	<0.001
Total Revenue	33.8M	85.6M	0	543M	489M	1,290M	-1.27M	8,800M	<0.001
Gross Margin (%)	45.874	28.349	-76.892	100.000	54.555	30.351	-76.892	100.000	0.002
Capital Expenditures	-2.37M	6.38M	-58.4M	0	-21.1M	76.6M	-597M	0	0.001
Total Assets	49.3M	86.1M	0.449M	718M	828M	1,990M	0.449M	12,300M	<0.001
Market Capitalization	44.4M	69.9M	0.818M	450M	647M	1,360M	0.818M	8,590M	<0.001
Number of Employees	182.952	359.065	0	2,750	3,229	10,251	0	70,376	<0.001
Board Size	4.989	1.477	2	10	6.263	1.976	2	12.5	<0.001
Ind. Directors Share	0.283	0.255	0	1	0.505	0.286	0	1	<0.001
Network Size	753.883	696.138	20.467	2,961	943.034	653.632	20.467	2,961	0.003
Directors' Age	56.181	5.611	43.667	69.4	56.723	4.557	43.667	69	0.247
Directors' Qualifications	1.457	0.673	0.3	3.2	1.619	0.578	0.3	3.2	0.006
Observations	189				287				

Notes: M = millions. p-values are from two-sample t-tests comparing treated and control groups.

Empirical Strategy

A simple comparison of average ROA between treated and control firms would produce biased estimates because treatment is not randomly assigned.

To estimate the causal effect of female board representation on firm performance, we rely on selection-on-observables methods. Under the Conditional Independence Assumption, treatment assignment is assumed to be independent of potential outcomes conditional on observed firm characteristics.

We begin by estimating the propensity score, defined as the probability that a firm appoints at least one female director after 2015. The propensity score is estimated using a logit model, where the dependent variable is an indicator equal to one if a firm is treated, and zero otherwise. All covariates are measured in 2010 to ensure they are predetermined with respect to treatment.

Table 2: Propensity Score Logit

Variable	Coefficient	Std. Error	Z-stat	p-value
ROA 2010	-0.007	0.004	-1.590	0.112
Total Revenue	<0.001	<0.001	0.724	0.469
Gross Margin (%)	0.005	0.005	1.008	0.313
Capital Expenditures	<0.001	<0.001	0.485	0.628
Total Assets	<0.001	<0.001	0.734	0.463
Market Capitalization	<0.001*	<0.001	2.508	0.012
Number of Employees	0.001	0.001	1.946	0.052
Board Size	0.106	0.101	1.057	0.291
Ind. Directors Share	0.305	0.567	0.537	0.591
Network Size	<0.001	0.001	1.294	0.196
Directors' Age	-0.012	0.028	-0.436	0.663
Directors' Qualifications	0.460*	0.228	2.016	0.044
Constant	-3.137	1.827	-1.717	0.086
Observations	437			

Notes: * $p < 0.05$. Only selected covariates from Table 1 included for clarity.

The estimated propensity scores are then used to match treated and control firms with similar observable characteristics. This allows us to compare firms that are comparable in terms of pre-treatment characteristics and isolate the effect of female board representation.

The results of the propensity score estimation are reported in Table 2. We find that market capitalization and directors' qualifications are significant predictors of treatment status: larger firms and firms with more qualified directors are more likely to appoint female board members.

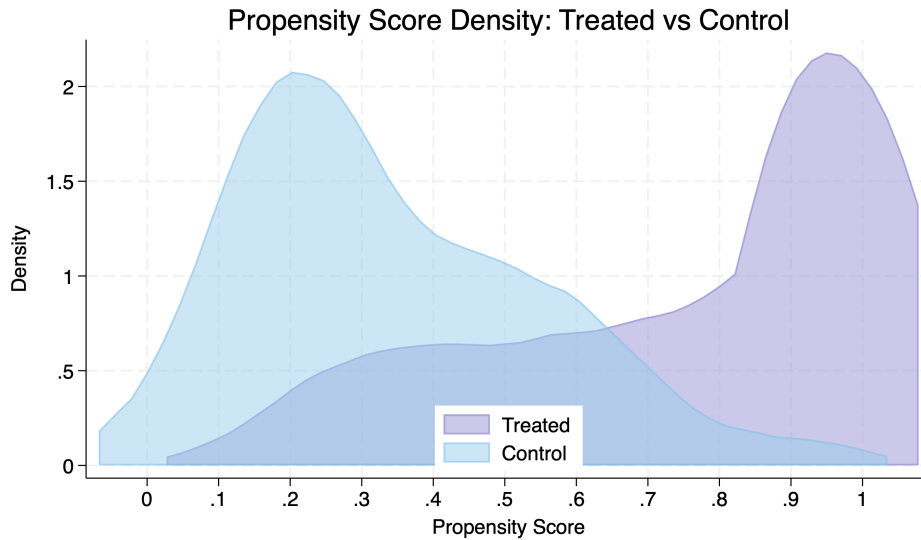


Figure 1: Distribution of propensity scores by treatment status

A key requirement for matching is the overlap assumption. To assess this, we compare the

distribution of propensity scores between treated and control firms using kernel density plots. Figure 1 shows substantial overlap between the two groups, indicating that for most treated firms there exist comparable control firms.

To further ensure comparability, we restrict the sample by trimming observations with extreme propensity scores (below 0.05 and above 0.95). This step reduces the influence of observations for which no suitable matches exist.

We then implement one-to-one nearest neighbor matching based on the propensity score. To evaluate the quality of matching, we examine covariate balance before and after matching. Table 3 reports standardized differences in covariates as well as t-tests for equality of means.

Before matching, treated and control firms differ substantially across most covariates, with average standardized bias around 30% and several variables exceeding 50%. After matching, the bias is substantially reduced: for most covariates it falls below 10–15%, and differences between groups are no longer statistically significant.

Table 3: Covariate Balance Before and After Matching

Variable	Means (Unmatched)		Standardized Bias (%)		p-value	
	Treated	Control	Before	After	Before	After
ROA 2010	-3.164	-9.855	20.0	1.5	0.074	0.894
Total Revenue	59.0M	30.0M	31.2	-11.2	0.005	0.441
Gross Margin (%)	55.106	46.355	29.2	19.9	0.009	0.101
Capital Expenditures	-3.4M	-2.2M	-16.7	-4.6	0.129	0.599
Total Assets	98.0M	46.0M	47.8	-14.7	<0.001	0.190
Market Capitalization	97.0M	41.0M	63.7	13.8	<0.001	0.178
Number of Employees	303.9	158.3	36.3	3.7	0.001	0.754
Board Size	5.474	4.959	33.7	-24.3	0.003	0.050
Ind. Directors Share	0.438	0.292	51.4	34.3	<0.001	0.003
Network Size	815.7	778.8	5.6	-51.2	0.615	<0.001
Directors' Age	56.536	56.090	8.4	0.7	0.453	0.951
Directors' Qualifications	1.552	1.458	14.5	-0.6	0.193	0.959

Notes: M = millions. Standardized bias and p-values are reported before and after propensity score matching.

These results indicate that matching significantly improves the comparability of treated and control firms. While some imbalance remains for certain variables (such as directors' network size), the overall balance is considerably improved. Figure A3 presents a loveplot illustrating the reduction in standardized bias across covariates after matching, confirming the effectiveness of the matching procedure.

Results

Table 4 presents the main estimates of the effect of female board representation on firm performance using several empirical approaches.

We begin with a naive OLS specification that regresses post-treatment ROA on the treatment indicator only. This specification suggests a statistically significant effect of female board

Table 4: Treatment Effect of Female Directors on Post-ROA (2015–2019)

	(1) Naive OLS	(2) OLS + Controls + FE	(3) IPWRA	(4) Nearest Neighbor	(5) CEM
Female	8.781*** (3.243)	3.527 (3.095)	2.338 (2.983)	-2.032 (3.478)	10.305 (10.104)
Observations	322	322	322	206	43
Baseline controls	No	Yes	Yes	Yes	Yes ^(a)
Industry fixed effects	No	Yes	Yes	No	No
Method	None	None	IPW + RA	NN weights	CEM weights

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. (a) CEM matches exactly on coarsened baseline (2010) covariates: ROA₂₀₁₀, gross margin, total assets, market capitalization, board size, and average director age. Controls are balanced through matching; no additional regression adjustment is applied. Nearest Neighbor = 1-to-1 matching with caliper 0.05 and common support (ATT). IPWRA = doubly robust estimator.

representation on firm performance. However, this estimate does not account for systematic differences between treated and control firms.

To alleviate this, we estimate an OLS model including a rich set of firm-level controls measured in 2010, as well as industry fixed effects. The estimated effect becomes statistically insignificant, indicating that the initial OLS result was driven by observable differences across firms.

To further address selection bias, we implement inverse probability weighting with regression adjustment (IPWRA), a doubly robust estimator that combines outcome regression with inverse-probability weighting (Wooldridge, 2010). The results remain statistically insignificant.

We then turn to nearest neighbor propensity score matching (1-to-1), restricting the sample to the region of common support and trimming extreme propensity score values. The matching procedure yields a sample of 206 firms. The estimated treatment effect in the matched sample is again statistically insignificant.

Finally, we estimate the effect using coarsened exact matching (CEM), following Iacus et al. (2012). CEM preprocesses the data by temporarily coarsening continuous covariates into intervals and then performing exact matching within these strata. Treated and control firms are compared only within strata where both groups are present, while observations that fall into strata containing only treated or only control firms are discarded.

This approach improves covariate balance without relying on correct specification of the propensity score model, but typically results in a smaller effective sample. In our application, the stricter matching criteria reduce the sample to 43 firms. Despite improved comparability between treated and control firms, the estimated treatment effect remains statistically insignificant.

This result is consistent with the findings from propensity score matching and IPWRA, suggesting that the absence of an effect is not driven by the particular matching method used.

Overall, while the naive OLS specification suggests a positive and significant relationship between female board representation and firm performance, this result disappears once we account for observable characteristics and ensure comparability between treated and control firms. Across all identification strategies, we find no evidence of a statistically significant effect.

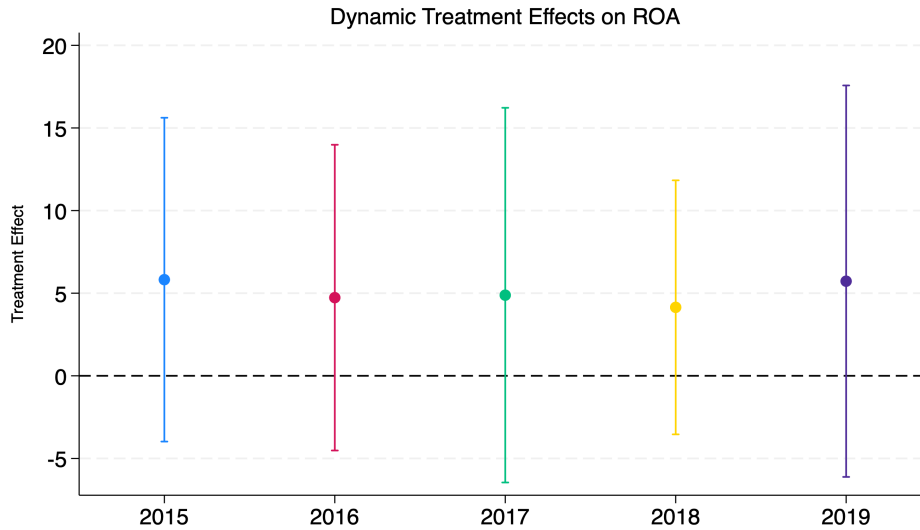


Figure 2: Dynamic effects of female board representation on ROA, 2015–2019

To complement the main results, we examine the dynamics of the treatment effect over time. Figure 2 plots year-by-year estimates of the effect of female board representation on ROA for the period 2015–2019, using inverse probability weighting. The estimates are statistically insignificant in each year, indicating that the absence of an effect is not driven by aggregation over time.

We also analyze the distribution of outcomes in the matched sample. Figure A4 shows kernel density plots of average ROA (2015–2019) for treated and control firms after matching. The distributions largely overlap, suggesting that the matching procedure successfully balances the two groups and that treated and control firms are comparable in terms of observable characteristics.

Robustness Check

Table 5 reports several robustness checks to assess the stability of the main results.

Table 5: Robustness Checks: Female on Board $\rightarrow$ ROA

	(1)	(2)	(3)	(4)
	Δ ROA	Placebo	Large firms	Small firms
Female on board	2.338	-4.762	-0.582	5.354
	(2.983)	(3.414)	(4.660)	(5.519)
Observations	322	322	161	161

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Column (1) reports the effect of a female director on Δ ROA (2015–2019 average minus 2010) using IPWRA with robust standard errors. Column (2) presents a placebo test using ROA in 2010. Columns (3) and (4) report results for large and small firms, defined by the median market capitalization. All specifications include baseline covariates and industry FE.

First, in column (1), we use an alternative outcome variable, defined as the change in ROA between 2010 and the average ROA over 2015–2019. This specification captures changes in firm

performance relative to the pre-treatment baseline. The estimate is obtained using IPWRA with the same set of firm-level controls measured in 2010 and industry fixed effects. The estimated treatment effect remains statistically insignificant.

Second, in column (2), we conduct a placebo test using ROA in 2010 as the dependent variable. Since this outcome is measured before the 2015 recommendation, any significant treatment effect would suggest pre-existing differences between treated and control firms and would cast doubt on the identifying assumptions. We estimate this specification using the same set of controls and industry fixed effects, excluding ROA in 2010 from the control set to avoid collinearity. The placebo coefficient is statistically insignificant, which supports the plausibility of our identification strategy.

Finally, in columns (3) and (4), we explore heterogeneity by firm size. We split firms into large and small groups based on the median market capitalization in the trimmed sample. We then estimate separate OLS regressions with the same controls and industry fixed effects for each subsample. The results remain statistically insignificant in both groups, suggesting that the absence of an effect is not driven by firm size.

Overall, the robustness checks confirm the main finding of the paper: once observable differences between firms are taken into account, there is no evidence that female board representation has a significant effect on firm performance.

Conclusion

This paper examines whether female representation on corporate boards affects firm performance. Using data on UK firms and the 2015 recommendation to increase female board representation, we estimate the effect of female directors on return on assets (ROA). To address non-random selection into treatment, we apply several selection-on-observables methods, including OLS with controls, propensity score matching, IPWRA, and CEM.

Across all specifications, we find no statistically significant effect of female board representation on firm performance. While a naive OLS regression suggests a positive relationship, this result disappears once observable firm characteristics are taken into account and treated firms are compared with similar control firms. The findings remain unchanged in robustness checks using alternative outcomes, placebo tests, and subsample analysis by firm size.

Overall, our results suggest that the positive correlation between gender-diverse boards and firm performance may reflect differences in firm characteristics rather than a causal effect of female directors. At the same time, the results should be interpreted cautiously, since the analysis relies on the Conditional Independence Assumption and the 2015 recommendation was non-binding. Future research may examine longer-run effects and alternative measures of firm performance.

A Additional Figures

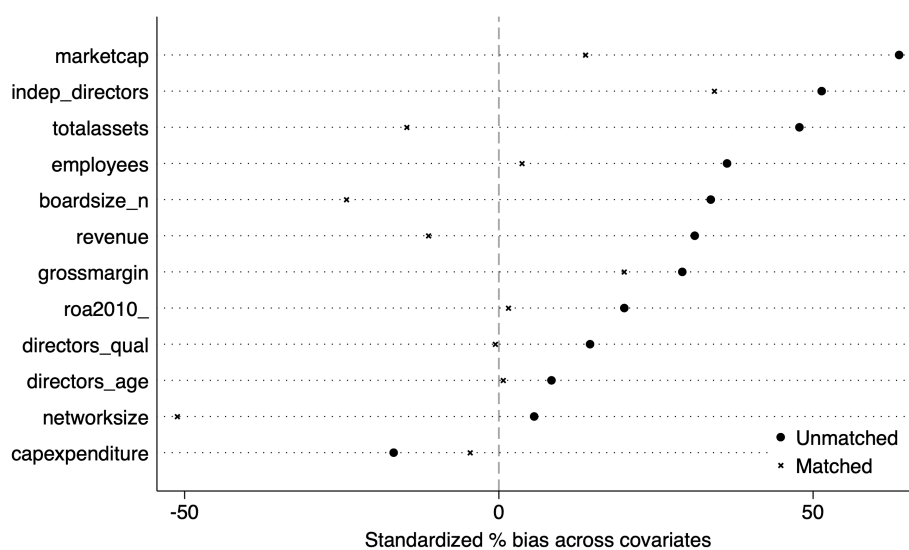


Figure A3: Covariate balance before and after matching

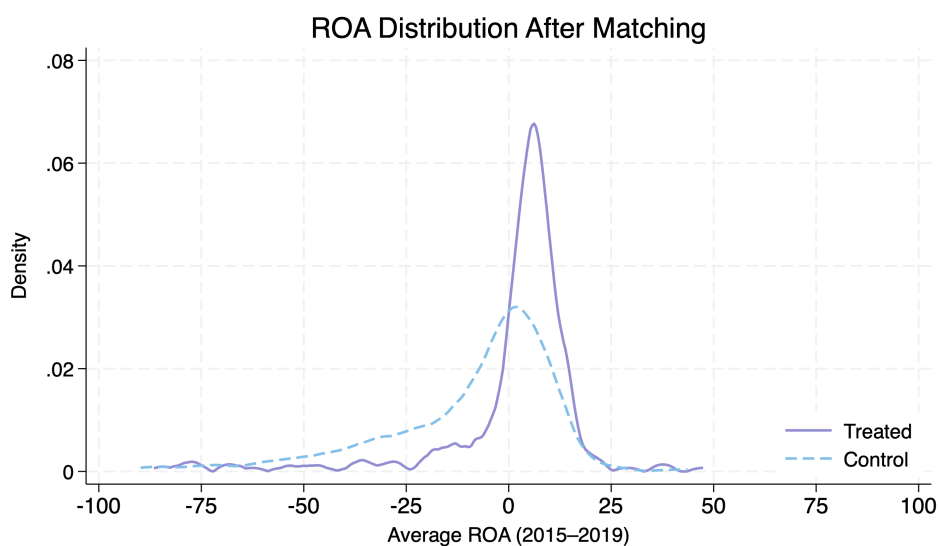


Figure A4: Distribution of ROA for treated and control firms after matching

References

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- Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data* (2 ed.). Cambridge, MA: MIT Press.